正交曲线坐标系的微分算符.

1. 正交曲线坐标系.

直角 (x, y, z). 球 (r, θ, φ) 柱 (ρ, ϕ, z).

正交曲线坐标系

一般: (q_1, q_2, q_3).

与直角坐标变换.

$$x = x(q_1, q_2, q_3),\quad y = y(q_1, q_2, q_3),\quad z = z(q_1, q_2, q_3)$$

一般要求 Jacobi 行列式

$$J = \frac{\partial(x, y, z)}{\partial(q_1, q_2, q_3)} = \begin{vmatrix} \partial x/\partial q_1 & \partial x/\partial q_2 & \partial x/\partial q_3 \\ \partial y/\partial q_1 & \partial y/\partial q_2 & \partial y/\partial q_3 \\ \partial z/\partial q_1 & \partial z/\partial q_2 & \partial z/\partial q_3 \end{vmatrix} \neq 0$$

$J = 0$ 处, 给定 (x, y, z) 或无法唯一确定 (q_1, q_2, q_3).

坐标轴的奇点.

Example: 直角 $\longrightarrow$ 球坐标系

$J = r^2 \sin\theta$, when $r = 0$ or $\theta = 0, \pi$, $J = 0$

在这些位置某些点无法定义.

但没有关系, 这些点构成的集合测度为 0.

类似体积.

$$x = x(q_1, q_2, q_3),\quad y = y(q_1, q_2, q_3),\quad z = z(q_1, q_2, q_3)$$

$$\downarrow$$

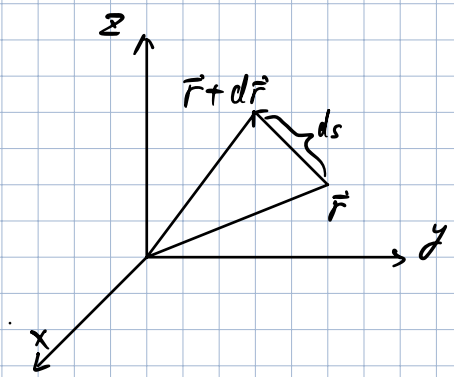
$$\vec{r} = \vec{r}(q_1, q_2, q_3)$$

Q: 什么曲线坐标系才正交?

考虑直角坐标.

Def: 距离 ds

$$(ds)^2 = d\vec{r}\cdot d\vec{r} = (dx)^2 + (dy)^2 + (dz)^2$$



Note:

- 上述定义 ⟶ 平直 Euclid 空间
- 其它空间定义不同.

一般曲线坐标:

$$d\vec{r} = \sum_{i=1}^{3} \frac{\partial \vec{r}}{\partial q_i} dq_i$$

$$(ds)^2 = d\vec{r}\cdot d\vec{r} = \sum_{i,j=1}^{3} \frac{\partial \vec{r}}{\partial q_i}\cdot\frac{\partial \vec{r}}{\partial q_j} dq_i\, dq_j$$

Def 正交

$$\frac{\partial \vec{r}}{\partial q_i}\cdot\frac{\partial \vec{r}}{\partial q_j} = 0,\quad i\neq j$$

交叉项为 0.

$$(ds)^2 = \sum_{i=1}^{3} \left(\frac{\partial \vec{r}}{\partial q_i}\right)^2 (dq_i)^2 \equiv \sum_{i=1}^{3} (h_i\, dq_i)^2$$

$h_i = \left|\frac{\partial \vec{r}}{\partial q_i}\right|$: 度规系数.

(拉梅系数).

Note: 度规: 度量长度的规则?

尺子的刻度?

空间变化后尺子发生变化, 需要建立某种联系.

单位矢量 $(\vec{e}_1, \vec{e}_2, \vec{e}_3)$?

$\vec{e}_1$ 沿 q_1 坐标线切线方向. 沿着 q_1.

$$d\vec{r} = \left(\frac{\partial \vec{r}}{\partial q_1}\right) dq_1 \,, \quad ds = |d\vec{r}| = h_1 dq_1$$

$\therefore$

$$\Rightarrow \vec{e}_1 = \frac{d\vec{r}}{ds} = \frac{1}{h_1}\left(\frac{\partial \vec{r}}{\partial q_1}\right)$$

矢量 / 矢量长度.

综上,

$$\vec{e}_i = \frac{1}{h_i}\left(\frac{\partial \vec{r}}{\partial q_i}\right), \quad i = 1, 2, 3.$$

由度规定义, 显然, $\vec{e}_i \cdot \vec{e}_j = 0, \; i \neq j$

Example 1. 柱坐标. (ρ, ϕ, z)

$$\begin{cases} x = \rho\cos\phi \\ y = \rho\sin\phi \\ z = z \end{cases} \qquad \vec{r} = \rho\cos\phi\, \hat{e}_x + \rho\sin\phi\, \hat{e}_y + z\hat{e}_z$$

度规: $h_i = \left|\frac{\partial \vec{r}}{\partial q_i}\right|$.

$$\begin{cases} h_1 = \left|\frac{\partial \vec{r}}{\partial \rho}\right| = |\cos\phi\, \hat{e}_x + \sin\phi\, \hat{e}_y| = 1 \\ h_2 = \left|\frac{\partial \vec{r}}{\partial \phi}\right| = |-\rho\sin\phi \hat{e}_x + \rho\cos\phi \hat{e}_y| = \rho \\ h_3 = \left|\frac{\partial \vec{r}}{\partial z}\right| = 1 \end{cases}$$

单位矢径

$$\begin{cases} \hat{e}_\rho = \frac{1}{h_1}\frac{\partial \vec{r}}{\partial \rho} = \cos\phi\, \hat{e}_x + \sin\phi\, \hat{e}_y \,, \\ \hat{e}_\phi = \frac{1}{h_2}\frac{\partial \vec{r}}{\partial \phi} = -\sin\phi\, \hat{e}_x + \cos\phi\, \hat{e}_y \\ \hat{e}_z = \hat{e}_z \end{cases}$$

弧长：由 $\vec{\phi} \to d\vec{\phi} + \vec{\phi}$

$$h_i\, d\vec{\phi} = \rho\, d\vec{\phi}$$

Example 2. 球坐标，(r, θ, φ).

$$\begin{cases} x = r\sin\theta\cos\varphi. \\ y = r\sin\theta\sin\varphi \\ z = r\cos\theta \end{cases} \Rightarrow \vec{r} = r\sin\theta\cos\varphi\,\hat{e}_x + r\sin\theta\sin\varphi\,\hat{e}_y + r\cos\theta\,\hat{e}_z$$

$\therefore$ 度规 $h_r = \left|\frac{\partial \vec{r}}{\partial r}\right| = \left|\sin\theta\cos\varphi\,\hat{e}_x + \sin\theta\sin\varphi\,\hat{e}_y + \cos\theta\,\hat{e}_z\right| = 1.$

$h_\theta = \left|\frac{\partial \vec{r}}{\partial \theta}\right| = \left|r\cos\theta\cos\varphi\,\hat{e}_x + r\cos\theta\sin\varphi\,\hat{e}_y - r\sin\theta\,\hat{e}_z\right| = 1.$

$h_\varphi = \left|\frac{\partial \vec{r}}{\partial \varphi}\right| = \left|-r\sin\theta\sin\varphi\,\hat{e}_x + r\sin\theta\cos\varphi\,\hat{e}_y\right| = r\sin\theta.$

单位矢.

$$\begin{cases} \hat{e}_r = \sin\theta\cos\varphi\,\hat{e}_x + \sin\theta\sin\varphi\,\hat{e}_y + \cos\theta\,\hat{e}_z \\ \hat{e}_\theta = r\cos\theta\cos\varphi\,\hat{e}_x + r\cos\theta\sin\varphi\,\hat{e}_y - r\sin\theta\,\hat{e}_z \\ \hat{e}_\varphi = (-r\sin\theta\sin\varphi\,\hat{e}_x + r\sin\theta\cos\varphi\,\hat{e}_y)/r\sin\theta = -\sin\varphi\,\hat{e}_x + \cos\varphi\,\hat{e}_y \end{cases}$$

正交曲线坐标系中微分算符.

微分算符： $\nabla(A)$. $\nabla\cdot A$. $\nabla\times A$. $\nabla^2 A. = \nabla\cdot(\nabla A)$

Q: 用度规求这些算符?

Example: ∇u.

∇u is an vector. $\nabla u = \sum_{i=1}^{3} f_i \hat{e}_i$

由正交： $f_i = \hat{e}_i \cdot (\nabla u)$

同样，线元 $d\vec{s} = \sum_{i=1}^{3} ds_i$， $ds_i = \hat{e}_i \cdot d\vec{s}$

由梯度定义　　标量，或矢量的"长度"随空间距离的变化

$$\hat{e}_i \cdot \nabla u = \frac{\partial u}{\partial s_i} = \lim_{\Delta s \to 0} \frac{\Delta u}{\Delta s_i} = \lim_{\Delta q_i \to 0} \frac{1}{h_i} \frac{\Delta u}{\Delta q_i} = \frac{1}{h_i} \frac{\partial u}{\partial q_i}$$

综上，

$$f_i = \frac{1}{h_i} \frac{\partial u}{\partial q_i}, \quad \nabla u = \sum_{i=1}^{3} \frac{1}{h_i} \frac{\partial u}{\partial q_i} \hat{e}_i$$

直角坐标系中，

$$\nabla u = \frac{\partial u}{\partial x} \hat{e}_x + \frac{\partial u}{\partial y} \hat{e}_y + \frac{\partial u}{\partial z} \hat{e}_z$$

由上面结论：

柱坐标：

$$\nabla u = \frac{\partial u}{\partial \rho} \hat{e}_\rho + \rho \frac{\partial u}{\partial \phi} \hat{e}_\phi + \frac{\partial u}{\partial z} \hat{e}_z$$

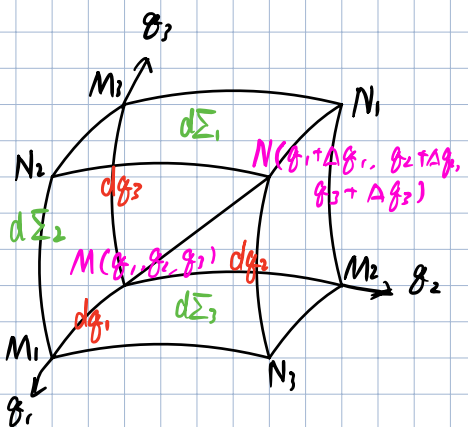
球坐标：

$$h_1=\left|\frac{\partial \vec{r}}{\partial r}\right|=1 \quad h_2=\left|\frac{\partial \vec{r}}{\partial \theta}\right|=r \quad h_3=r\sin\theta$$

$$\nabla u=\frac{\partial u}{\partial \rho}\hat{e}_\rho+\frac{1}{r}\frac{\partial u}{\partial \theta}\hat{e}_\theta+\frac{1}{r\sin\theta}\frac{\partial u}{\partial \varphi}\hat{e}_\varphi$$

散度推导

$$\nabla\cdot\vec{A}=\lim_{\Delta V\to 0}\frac{\oint_{\partial(\Delta V)}\vec{A}\cdot d\vec{S}}{\Delta V}$$



- $\oint_{\partial(\Delta V)}\vec{A}\cdot d\vec{S}$ 在一个高斯面上积分.
- 高斯面所围成体积 $\longrightarrow 0$.

高斯面是任意的，故可取右图所示六面体.（近似长方体）

g_1，$g_1+\Delta g_1$；g_2，$g_2+\Delta g_2$；g_3，$g_3+\Delta g_3$

体积元 $\Delta V=(h_1\Delta g_1)\cdot(h_2\Delta g_2)\cdot(h_3\Delta g_3)$

$$\begin{aligned}\oint_{\partial(\Delta V)}\vec{A}\cdot d\vec{S}=&\left[(A_1h_2h_3)_{g_1+\Delta g_1}-(A_1h_2h_3)_{g_1}\right]\Delta g_2\Delta g_3 \quad \frac{\Delta g_1}{\Delta g_1} \quad \frac{\partial(A_1h_2h_3)}{\partial g_1}\\ &+\left[(A_2h_1h_3)_{g_2+\Delta g_2}-(A_2h_1h_3)_{g_2}\right]\Delta g_1\Delta g_3 \quad \frac{\Delta g_2}{\Delta g_2} \quad \frac{\partial(A_2h_1h_3)}{\partial g_2}\\ &+\left[(A_3h_1h_2)_{g_3+\Delta g_3}-(A_3h_1h_2)_{g_3}\right]\Delta g_1\Delta g_2 \quad \frac{\Delta g_3}{\Delta g_3} \quad \frac{\partial(A_3h_1h_2)}{\partial g_3}\\ =&\left[\frac{\partial(A_1h_2h_3)}{\partial g_1}\Big|_{r_0}+\frac{\partial(A_2h_1h_3)}{\partial g_2}\Big|_{r_0}+\frac{\partial(A_3h_1h_2)}{\partial g_3}\Big|_{r_0}\right]\Delta g_1\Delta g_2\Delta g_3\end{aligned}$$

∴最终，$$\nabla\cdot\vec{A}=\frac{1}{h_1h_2h_3}\left[\frac{\partial(A_1h_2h_3)}{\partial q_1}+\frac{\partial(A_2h_1h_3)}{\partial q_2}+\frac{\partial(A_3h_1h_2)}{\partial q_3}\right].$$

Example 1 柱坐标

$$h_1=1,\quad h_2=r,\quad h_3=1$$

$$\nabla\cdot A=\frac{1}{r}\left[\frac{\partial A_r}{\partial r}+\frac{\partial(rA_\phi)}{\partial\phi}+\frac{\partial A_z}{\partial z}\right]$$

Example 2. 球坐标.

$$h_1=1,\quad h_2=r,\quad h_3=r\sin\theta.$$

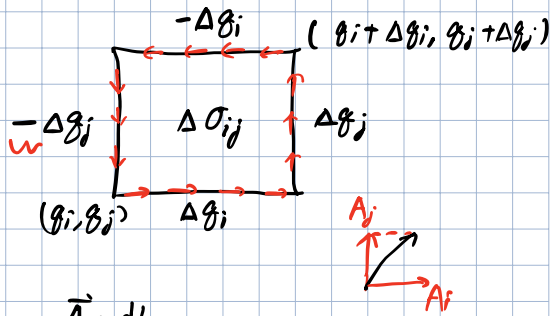
$$\nabla\cdot\vec{A}=\frac{1}{r^2\sin\theta}\left[\frac{\partial A_r}{\partial r}+\frac{\partial(rA_\theta)}{\partial\theta}+\frac{\partial(r\sin\theta A_\varphi)}{\partial\varphi}\right].$$

可由 $\nabla^2u=\nabla\cdot(\nabla A)$ 得：∇^2A.

$$\nabla^2A=\nabla\cdot(\nabla u)=\nabla\cdot\left[\sum_{i=1}^{3}\frac{1}{h_i}\left(\frac{\partial u}{\partial q_i}\right)\hat{e}_i\right]$$

$$\nabla^2A=\frac{1}{h_1h_2h_3}\left\{\frac{\partial}{\partial q_1}\left[\frac{h_2h_3}{h_1}\left(\frac{\partial u}{\partial q_1}\right)\right]+\frac{\partial}{\partial q_2}\left[\frac{h_1h_3}{h_2}\left(\frac{\partial u}{\partial q_2}\right)\right]+\frac{\partial}{\partial q_3}\left[\frac{h_1h_2}{h_3}\left(\frac{\partial u}{\partial q_3}\right)\right]\right\}$$

旋度 $\nabla \times \vec{A} = \lim_{\Delta\sigma \to 0} \frac{\oint_{\partial(\Delta\sigma)} \vec{A} \cdot d\vec{l}}{\Delta\sigma}$



含其沿 $\hat{e}_1, \hat{e}_2, \hat{e}_3$ 分解.

$$\nabla \times \vec{A} = \sum_{i,j} (\nabla \times A)_{i,j} \hat{e}_k$$

$$(\nabla \times \vec{A})_k = \lim_{\Delta\sigma_{ij} \to 0} \frac{\oint_{\partial(\Delta\sigma_{ij})} \vec{A} \cdot dL}{\Delta\sigma_{ij}}$$

$$\oint_{\partial(\Delta\sigma_{ij})} \vec{A} \cdot d\vec{l} = [-(A_i h_i)_{q_i+\Delta q_i} + (A_i h_i)_{q_i}]\Delta q_j + [(A_j h_j)_{q_j+\Delta q_j} - (A_j h_j)_{q_j}]\Delta q_i$$

$$= [-\frac{\partial(A_i h_i)}{\partial q_i} + \frac{\partial(A_j h_j)}{\partial q_j}]\Delta q_i \Delta q_j = \frac{1}{h_i h_j}[\frac{\partial(A_i h_i)}{\partial q_i} + \frac{\partial(A_j h_j)}{\partial q_j}]\Delta\sigma_{ij}$$

$\therefore$

$$(\nabla \times \vec{A})_k = \frac{1}{h_i h_j}[-\frac{\partial(A_i h_i)}{\partial q_i} + \frac{\partial(A_j h_j)}{\partial q_j}]$$

综上,

$$\begin{cases} (\nabla \times \vec{A})_1 = \frac{1}{h_2 h_3}[-\frac{\partial(A_2 h_2)}{\partial q_2} + \frac{\partial(A_3 h_3)}{\partial q_3}] \\ (\nabla \times \vec{A})_2 = \frac{1}{h_1 h_3}[-\frac{\partial(A_3 h_3)}{\partial q_3} + \frac{\partial(A_1 h_1)}{\partial q_1}] \\ (\nabla \times \vec{A})_3 = \frac{1}{h_1 h_2}[-\frac{\partial(A_1 h_1)}{\partial q_1} + \frac{\partial(A_2 h_2)}{\partial q_2}] \end{cases}$$

总结：

1. 度规系数（拉梅系数）

$$h_i = \left| \frac{\partial \vec{r}}{\partial q_i} \right| \quad , \quad i=1,2,3.$$

2. 向量 $\vec{r}$ 与 $\vec{r}+d\vec{r}$ 间距离 ds

$$(ds)^2 = \sum_{i=1}^{3} (h_i\, dq_i)^2$$

3. 单位向量.

$$\hat{e}_i = \frac{1}{h_i} \frac{\partial \vec{r}}{\partial q_i} \quad , \quad i=1,2,3.$$

4. 梯度

$$\nabla u = \sum_{i=1}^{3} \frac{1}{h_i} \frac{\partial u}{\partial q_i}$$

5. 散度

$$\nabla \cdot \vec{A} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial (h_2 h_3 A_1)}{\partial q_1} + \frac{\partial (h_1 h_3 A_2)}{\partial q_2} + \frac{\partial (h_1 h_2 A_3)}{\partial q_3} \right].$$

6. Laplace 算符：

$$\nabla^2 A = \nabla \cdot (\nabla A) = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial u}{\partial q_1} \right) + \frac{\partial}{\partial q_2} \left(\frac{h_1 h_3}{h_2} \frac{\partial u}{\partial q_2} \right) + \frac{\partial}{\partial q_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial u}{\partial q_3} \right) \right]$$

7. 旋度.

$$(\nabla \times \vec{A})_1 = \frac{1}{h_2 h_3} \left[- \frac{\partial (A_2 h_2)}{\partial q_2} + \frac{\partial (A_3 h_3)}{\partial q_3} \right]$$

$$(\nabla \times \vec{A})_2 = \frac{1}{h_1 h_3} \left[- \frac{\partial (A_3 h_3)}{\partial q_3} + \frac{\partial (A_1 h_1)}{\partial q_1} \right]$$

$$(\nabla \times \vec{A})_3 = \frac{1}{h_1 h_2} \left[- \frac{\partial (A_1 h_1)}{\partial q_1} + \frac{\partial (A_2 h_2)}{\partial q_2} \right].$$